

Andy Dam

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EDUCATION

The University of Texas at Dallas

Master of Science in Computer Science | GPA: 4.000

Bachelor of Science in Computer Science | GPA: 3.984

Richardson, TX

May 2027

May 2026

EXPERIENCE

Visa

May 2026 – Aug 2026

Software Engineering Intern

Highlands Ranch, CO

- Architected a full-stack regression-triage dashboard using **React**, **Vite**, **Node.js**, and **Express**, replacing manual **Jenkins** and test-report investigation with a unified view of failure verdicts, trends, and recommended actions
- Built a two-stage **LLM** pipeline with a shared cache contract, keeping per-run summarization and cross-run root-cause analysis decoupled and independently triggerable with live decision-stream telemetry streaming
- Automated a previously manual workflow spanning **Jenkins**, **Artifactory**, CLI tools, and test reports, with an estimated annual impact of approximately **1,000 engineering hours** reclaimed
- Conducted project demonstrations during sprint review meetings for **30+ engineers** and incorporated 5+ requested enhancements from direct feedback of our consumer base

Cavall Labs — CMU VentureBridge

Jan 2026 – May 2026

Software Engineering Intern

Remote

- Architected a RESTful API backend in **FastAPI** to power an AI-driven molecular energy prediction engine platform for molecular reactions, molecule bond strength estimations, and molecule image parsing in **Next.js**
- Engineered a background worker process system to decouple intensive GPU calculations away from the **FastAPI** backend while utilizing **PostgreSQL** as a job queue store
- Built a molecule-level and reaction-level caching layer with **PostgreSQL** to avoid redundant chemistry computations, tracking molecule status across pending, processing, completed, and error states

Paycom

May 2025 – Aug 2025

Software Development Intern

Irving, TX

- Engineered a full-stack anonymous survey application in **React** to internalize third-party vendor services and secure sensitive employee feedback data, deployed to **37,500+ clients** on the main commercial platform
- Architected scalable backend of **20+ RESTful endpoints** in **C#/.NET Core** and leveraged **Dapper** ORM to optimize **MySQL** data mapping for significantly reduced query response times

PROJECTS

Tracen Replay | Go, Python, Vue.js, TypeScript, PyTorch, PostgreSQL, Azure

- Architected a full-stack video analysis platform with a **Go** backend, **Vue.js** client, and **Python** worker that converts *Umamusume: Pretty Derby* gameplay recordings into turn-by-turn reports synced to the source video
- Engineered an **OCR** pipeline with **ONNX Runtime**, parallel worker pools and targeted dense rereads of fast animations, cutting a full-career analysis from several CPU hours to **under an hour** on a GPU
- Designed a self-verifying accounting engine that reconciles every turn's stat changes against the next turn's observed values, flagging unexplained discrepancies for review and user correction
- Trained a custom **CNN** with **CTC** decoding from scratch in **PyTorch** on crops labeled automatically by the accounting engine, raising stat reads from **72% to 85%** over the existing OCR reader
- Shipped both a hosted service on **Azure Container Apps** and a cross-platform **Wails** desktop app for Windows and macOS that keeps recordings on the user's own machine and analyzes them on their own GPU

Song Mood Classifier | Python, pandas, scikit-learn, Flask, HTML, CSS

- Trained a **Random Forest** model on 278k songs using 9 audio features to classify 4 mood categories with **92% validation accuracy** after rigorous hyperparameter tuning
- Integrated **Spotify Web API** to retrieve real-time audio metrics including danceability, energy, valence, and acousticness, powering instant song classification through an interactive **Flask** web application

TECHNICAL SKILLS

Languages: Python, Go, JavaScript/TypeScript, C#, Java, C/C++, HTML/CSS, SQL

Frameworks/Libraries: React, Vue.js, .NET Core, FastAPI, PyTorch, pandas, Spark, Kafka, scikit-learn, Karate

Developer Tools: Git, Docker, Azure, GitHub Actions, VS Code, Claude Code, Cursor, Jupyter Notebook